

August 7, 2025

Energy Division
California Public Utilities Commission
505 Van Ness Avenue
San Francisco, California 94102
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RE: Response of the Vehicle-Grid Integration Council to Advice Letter 5585-E of Southern California Edison Company, Advice Letter 7651-E of Pacific Gas and Electric Company, Advice Letter 4694-E of San Diego Gas and Electric Company

Dear Sir or Madam:

Pursuant to the provisions of General Order 96-B, the Vehicle-Grid Integration Council (VGIC) hereby submits these comments to Advice Letter 5585-E of Southern California Edison (SCE) Company, Advice Letter 7651-E of Pacific Gas and Electric Company, (PG&E), and Advice Letter 4694-E of San Diego Gas and Electric Company (SDG&E), *Proposed Enhancement to the Vehicle-To-Grid (V2G) AC Pilot in Accordance with Resolution E-5315* (Joint Advice Letter) submitted on July 18, 2025.

Overall, VGIC supports the Advice Letter implementing an additional temporary pathway for vehicle-to-grid (V2G) alternating current (AC) interconnection. Utilizing the forthcoming UL 1741 Certification Requirement Decision (CRD) for DER Systems will allow additional V2G AC systems to interconnect in California. We also recommend that the Joint Investor-Owned Utilities (IOUs) consider whether the UL 1741 CRD pathway is appropriate for permanent adoption in Rule 21, alongside the UL 1741 SC pathway, once the V2G AC Interconnection Technical Subgroup is reconvened.

I. VGIC SUPPORTS THE APPROVAL OF UL 1741 CRD AS AN INTERCONNECTION PATHWAY FOR V2G AC WITHIN THE PILOT.

VGIC supports the IOUs' proposal to establish an alternative certification pathway for V2G AC interconnection using the forthcoming UL 1741 CRD for DER Systems, in combination with UL 1741 SB and CSIP. Historically, the V2G AC interconnection pilot has required the use of external relays. This approach imposes significant additional equipment cost as well as increasing overall complexity, requiring engineering expertise and added utility review of individual project configurations.

In contrast, the proposed CRD-based certification pathway offers a more cost-effective solution. Importantly, VGIC posits this approach would streamline interconnection review for utilities; standardized testing and certification to the CRD reduce the need for detailed project-by-project engineering assessments, enabling faster approvals and lowering administrative workload and costs.

VGIC also supports the Utilities' proposal to exclude V2G AC systems certified under this alternative process from the original V2G AC Pilot project cap of six projects per utility per year. Based on our understanding, the six-project cap was adopted as a tool for cost containment. However, as noted above, engineering review and administrative costs are anticipated to be lower under the CRD-based pathway compared to relay-based site designs. Eliminating the six-project cap ensures V2G AC deployments are not unduly constrained, allowing California to accelerate the adoption of bidirectional EV charging solutions that can relieve distribution capacity constraints, generator capacity constraints, generation energy costs, and related grid balancing costs and, in turn, upward pressure on rates. VGIC therefore urges the Commission to approve this proposed enhancement to the V2G AC Pilot.

II. THE UTILITIES SHOULD CONSIDER INCORPORATING THE UL 1741 CRD FOR DER SYSTEMS AS A PERMANENT CERTIFICATION PATHWAY ONCE THE V2G AC SUBGROUP IS RECONVENED.

VGIC supports the utilities' near-term implementation of UL 1741 CRD as an optional certification pathway within the V2G AC Pilot, but the Joint IOUs, alongside the other participants of the V2G AC Interconnection Technical Subgroup, once it reconvenes, should consider how this certification could be used going forward. Currently, V2G AC interconnection is only allowed under the pilot and has not been incorporated into Rule 21 on a permanent basis. The Advice Letter notes that the Joint IOUs are awaiting finalization of UL 1741 SC to establish permanent pathways, referring to it as "the leading standard for the interconnection of V2G AC systems."

VGIC does not believe V2G AC interconnection should be unduly restricted, including by limiting eligible interconnection pathways. Specifically, stakeholders should consider the merits of establishing multiple pathways within Rule 21 for enabling V2G AC interconnection. Both UL 1741 SC and UL 1741 CRD for DER Systems are designed to meet the critical underlying safety and grid interoperability requirements. Note UL 1741 CRD is for "matched pair" certification of a given EVSE paired with a specific vendor EV for V2G AC operation, whereas the UL 1741 SC certification is for general use with any EV meeting SAE J3072 certification. There is also a certification pathway for "matched pair" V2G AC using UL 9741 for grid-tied AC power export. The UL 9741 certification pathway, however, is not referenced in Rule 21, hence the desire to establish the UL 1741 CRD alternative certification pathway. Given the relatively nascent stage of

the market, VGIC recommends considering allowing customers to elect the interconnection pathways best suited for them and their choice of technology. Notably, Maryland has already established a dual pathway approach, and other markets, including Texas and Nevada, are faced with a timely opportunity to follow suit.

While permanent changes to Rule 21 are outside the scope of this Advice Letter, VGIC urges the Joint IOUs not to create undue limitations at this stage. Once the V2G AC Subgroup reconvenes, a robust stakeholder process should consider all reasonable certification pathways.

III. CONCLUSION.

VGIC appreciates the opportunity to submit these comments on the Joint Advice Letter. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

**VEHICLE-GRID INTEGRATION
COUNCIL**

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